Forecasting

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FACULTY OF
BUSINESS ECONOMICS

Outline

- Why do we need forecasting?
- Characteristics of forecasts
- Time series forecasting methods
- Measures of forecast error

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Why do we need Forecasting?

- Forecasting = development of predictions (demand, interest rates, stock market,…) for the future

- The basis for all strategic and planning decisions in a supply chain
  - Marketing: sales force allocation, promotions, new product introduction
  - Finance: plant/equipment investment, budgetary planning
  - Personnel: workforce planning, hiring, layoffs
  - Production: capacity planning, inventory, aggregate planning

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Characteristics of Forecasts

- Forecasts are always wrong
  - Should include measure of error, reflecting accuracy of the forecast
- Long-term forecasts are less accurate than short-term forecasts
  - Try to reduce forecast horizon
  - Depends on your planning horizon! Forecasts are necessary to support planning
- Forecasting is most difficult for products with high demand uncertainty
  - Products in early stage of life cycle
  - Products with short life cycles
  - …

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Characteristics of forecasts

Low implied demand uncertainty

High implied demand uncertainty

"Functional products"

- Easy to forecast demand (low forecast error)
- Easy to match supply and demand
  - Low stockout rate
  - Low risk of leftovers
- Product margin low

Example: commodity products (salt, pasta,…)

"Innovative products"

- Hard to forecast demand (high forecast error)
- Hard to match supply and demand
  - High stockout rate
  - High risk of leftovers
- High product margins

Example: products with short life cycle, new products (high tech)

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Characteristics of Forecasts

- Aggregate forecasts are more accurate than disaggregate forecasts (smaller forecast error)
  - The higher the level of aggregation of a variable (eg, product family demand instead of end item level demand, demand per month instead of per week), the lower its variability.

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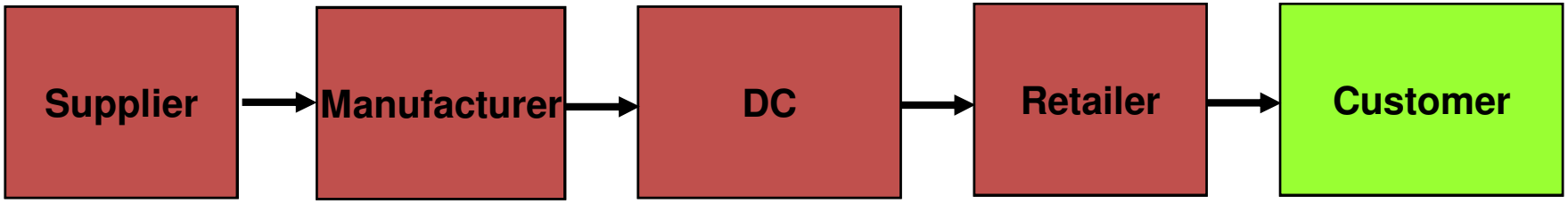
Characteristics of Forecasts

- Quality of the forecast depends on quality of the data on which it is based
  - Firms that are more UPSTREAM in the supply chain: tend to have larger forecast errors because they do not have direct view on end customer demand
  - Often, they just observe the replenishment orders from the next downstream stage and use this to develop forecasts
  - This is suboptimal: solution = collaborative forcasting: share info on en customer demand



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Forecasting Methods

- Bottomline:
  - Demand does not arise in a vacuum: influenced by a variety of **factors that are either known or can be predicted with some probability**
  - E.g.:
    - Past demand
    - Planned advertising efforts
    - Planned price discounts
    - …
- Expert judgment: when little or no data available
- Statistical forecasting methods: historical data available

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Forecasting Methods

- This course: Time-series methods:
  - Most appropriate when future demand is related to historical demand
  - <u>BUT:</u> demand ALWAYS contains a RANDOM component that cannot be explained by historical patterns

- Example: Tahoe Salt:
  - Produces and sells rock salt to melt snow

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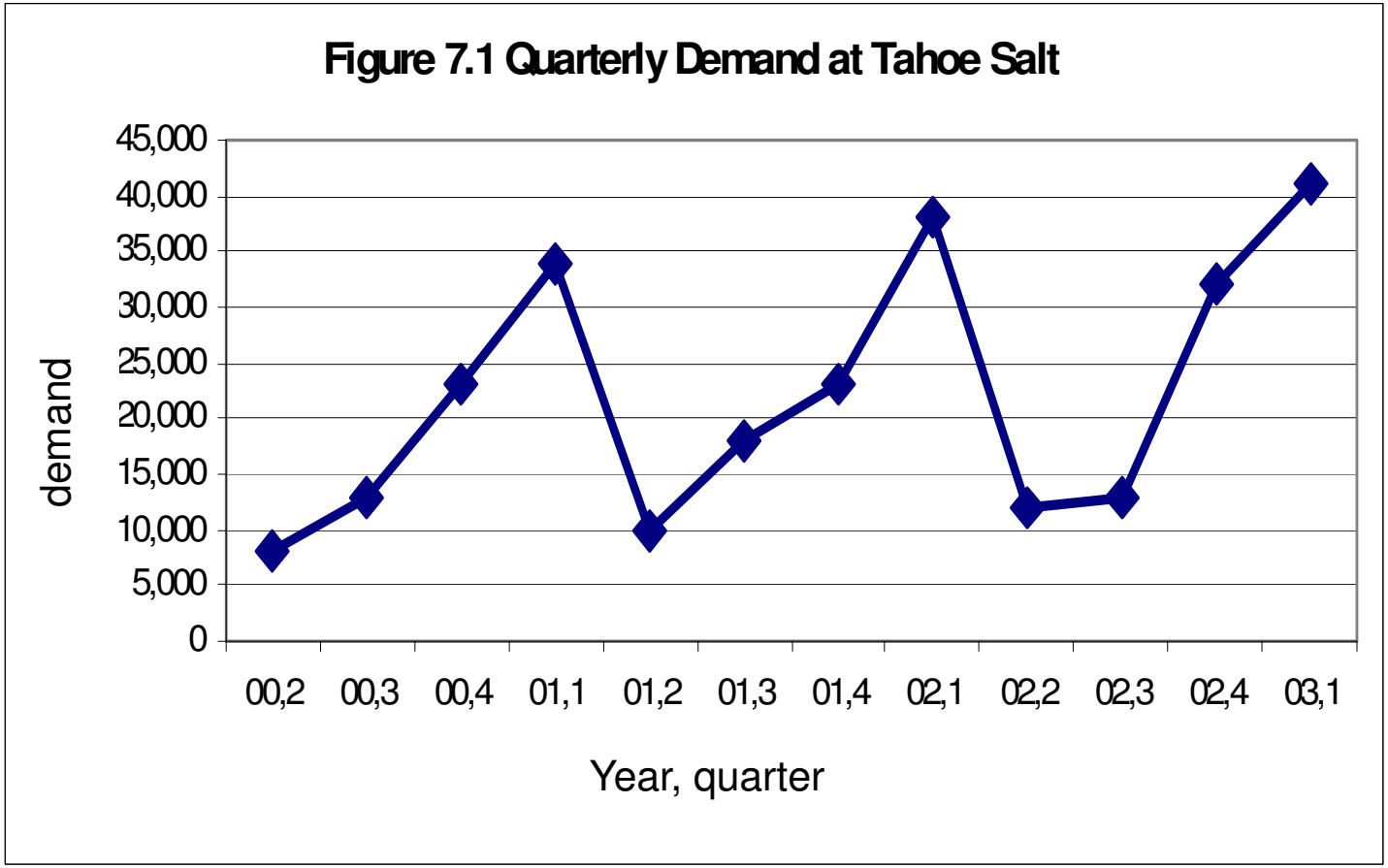
Forecasting Methods



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Components of an Observation

Observed demand (O) =
Systematic component (S) + Random component (R)

- ***Level*** (baseline)
- ***Trend*** (rate of growth/decline per period)
- ***Seasonality*** (predictable seasonal fluctuation)

- Systematic component: Expected value of demand:
  - Can be forecasted
- Random component (*noise*): The part of the demand that deviates from the systematic component
  - Cannot (and should not) be forecasted → shows up in forecast error
- Forecast error: difference between forecast and actual demand

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Time Series Forecasting Methods

- Goal is to predict **systematic component** of demand

Mixed: (level + trend)(seasonal factor)

- Static method:
  - Assumes that estimates of level, trend and seasonality DO NOT change as new demand is observed

- Adaptive methods:
  - Estimates of level, trend and seasonality are UPDATED after each new demand observation
    - Moving average **(no trend, no seasonality)**
    - Simple exponential smoothing **(no trend, no seasonality)**
    - Holt's model (**with trend**)
    - Winter's model (**with trend and seasonality**)

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Time Series Forecasting Methods

Does this mean that forecast error is not important?

- No!
- We just cannot FORECAST the error, we can only MEASURE it based on past performance

See later!

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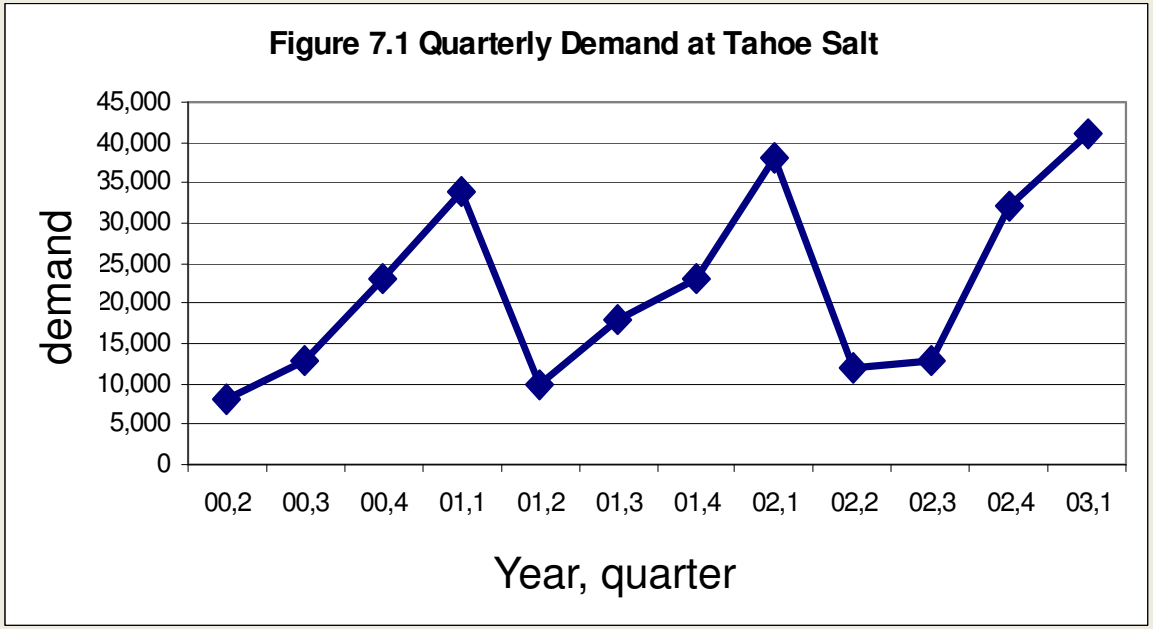
Time Series Forecasting: Tahoe Salt

Quarter	Demand D_t
II, 1998	8000
III, 1998	13000
IV, 1998	23000
I, 1999	34000
II, 1999	10000
III, 1999	18000
IV, 1999	23000
I, 2000	38000
II, 2000	12000
III, 2000	13000
IV, 2000	32000
I, 2001	41000

<u>Given:</u> Observations of past demand for the last 12 quarters.
<u>Question:</u> Forecast the demand for the next four quarters.



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Static forecasting

Static Method

- Assume a mixed model:

Systematic component = (level + trend)(seasonal factor)

$$F_k = [L + kT]S_i$$
$$= \text{forecast for demand in period } k$$

L = estimate of level at t= 0

T = estimate of trend (rate of increase/decrease in demand per period)

S_i = estimate of seasonal factor for season *i* to which period *k* belongs

D_k = actual demand in period *k*

F_k = forecast of demand in period *k*

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Estimating Level and Trend

- If you believe there is a seasonal effect in the data, you should deseasonalize demand before estimating level and trend
  - Deseasonalized demand = demand that would have been observed in the absence of seasonal fluctuations
  - Deseasonalized demand: contains only level and trend

- How? Determine periodicity (p)
  - the number of periods after which the seasonal cycle repeats itself
  - for demand at Tahoe Salt: $p = 4$
  - Calculate deseasonalized demand in period t as an average demand over p consecutive periods

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How to determine periodicity p from the data?

Determine periodicity p: calculate correlation coefficient ρ for different time lags (lag = 1 period, 2 periods, …)

Function = CORREL(array1,array2) in Excel

Year, Qtr	Period t	Demand D_t	lag 1	lag 2	lag 3	lag 4
00,2	1	8,000				
00,3	2	13,000	8,000			
00,4	3	23,000	13,000	8,000		
01,1	4	34,000	23,000	13,000	8,000	
01,2	5	10,000	34,000	23,000	13,000	8,000
01,3	6	18,000	10,000	34,000	23,000	13,000
01,4	7	23,000	18,000	10,000	34,000	23,000
02,1	8	38,000	23,000	18,000	10,000	34,000
02,2	9	12,000	38,000	23,000	18,000	10,000
02,3	10	13,000	12,000	38,000	23,000	18,000
02,4	11	32,000	13,000	12,000	38,000	23,000
03,1	12	41,000	32,000	13,000	12,000	38,000
			41,000	32,000	13,000	12,000
				41,000	32,000	13,000
					41,000	32,000
						41,000
		correlation ρ	0.085	-0.593	-0.192	0.945

So periodicity = 4

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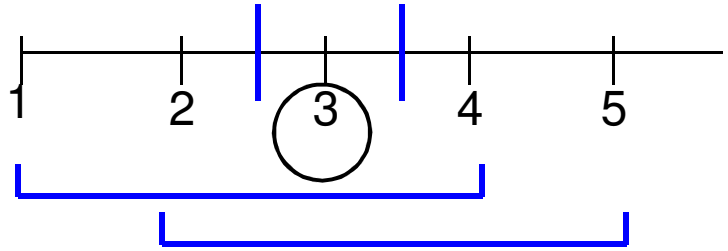
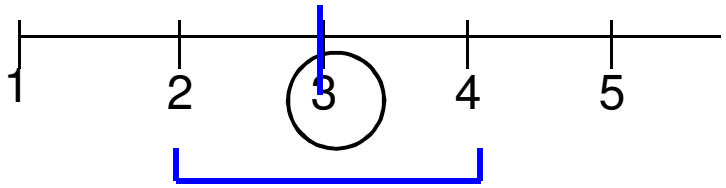
Estimating Level and Trend

- How to calculate deseasonalized demand for a period *t*?
  - Principle: Average demand from period i+1 to i+p provides deseasonalized demand for period i + (p+1)/2
  - Suppose periodicity *p* = even: e.g. p=4



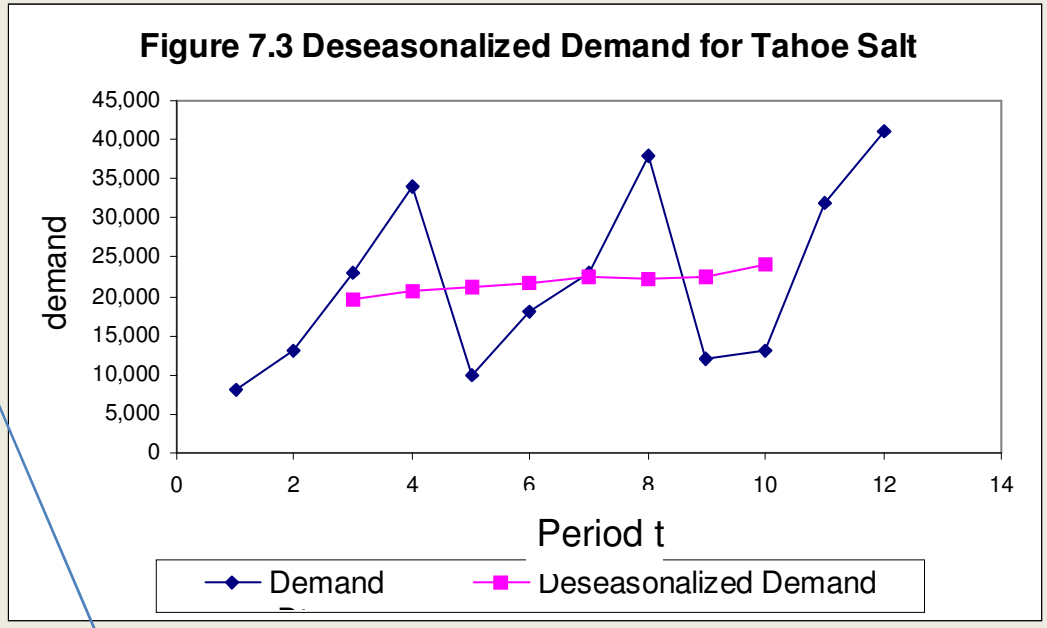
  - Suppose periodicity *p* = odd: e.g. p=3



Estimating Level and Trend

- Deseasonalizing demand for Tahoe Salt

Period t	*Demand* D_t	*Deseasonalized Demand*
1	8,000	
2	13,000	
3	23,000	19,750
4	34,000	20,625
5	10,000	21,250
6	18,000	21,750
7	23,000	22,500
8	38,000	22,125
9	12,000	22,625
10	13,000	24,125
11	32,000	
12	41,000	



(19500+20000)/2

Estimating level and trend

- From deseasonalized demand, obtain level and trend using following linear relationship:

$$\overline{D_t} = L + tT + \varepsilon_t$$

L = Level at period 0

ε_t = error (residual)

T = Trend: rate of growth/decline per period

  - Use linear regression:
    - » deseasonalized demand = dependent variable
    - » *t* = independent variable

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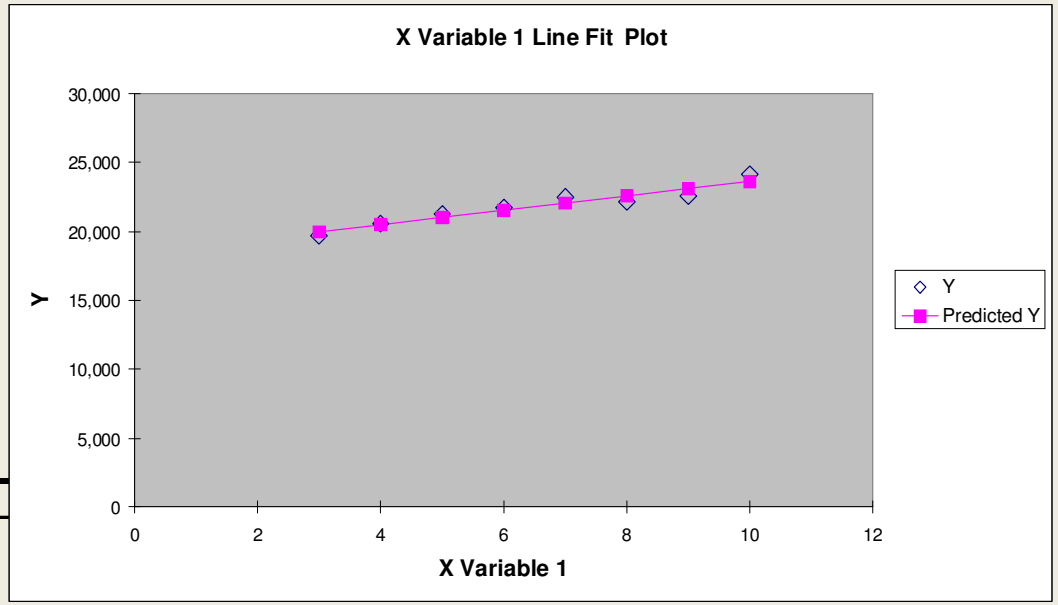
Estimating Level and Trend

Output of linear regression in Excel (Tools > Data analysis)



REGRESSION SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.958065237
R Square	0.917888998
Adjusted R Sqı	0.90420383
Standard Error	414.5033124
Observations	8

ANOVA

	df	*SS*	*MS*
Regression	1	11523809.52	11523810
Residual	6	1030877.976	171813
Total	7	12554687.5	

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	***18,439***	440.8087079	41.82991	1.25E-08	17360.36726	19517.609	17360.3673	19517.6089
X Variable 1	***524***	63.95924968	8.189738	0.000179	367.3067633	680.31228	367.306763	680.312284

Initial Level, L

Trend, T

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Estimating Level and Trend

- Once you have L and T, you can obtain <u>estimated deseasonalized demand</u> for all periods of past demand

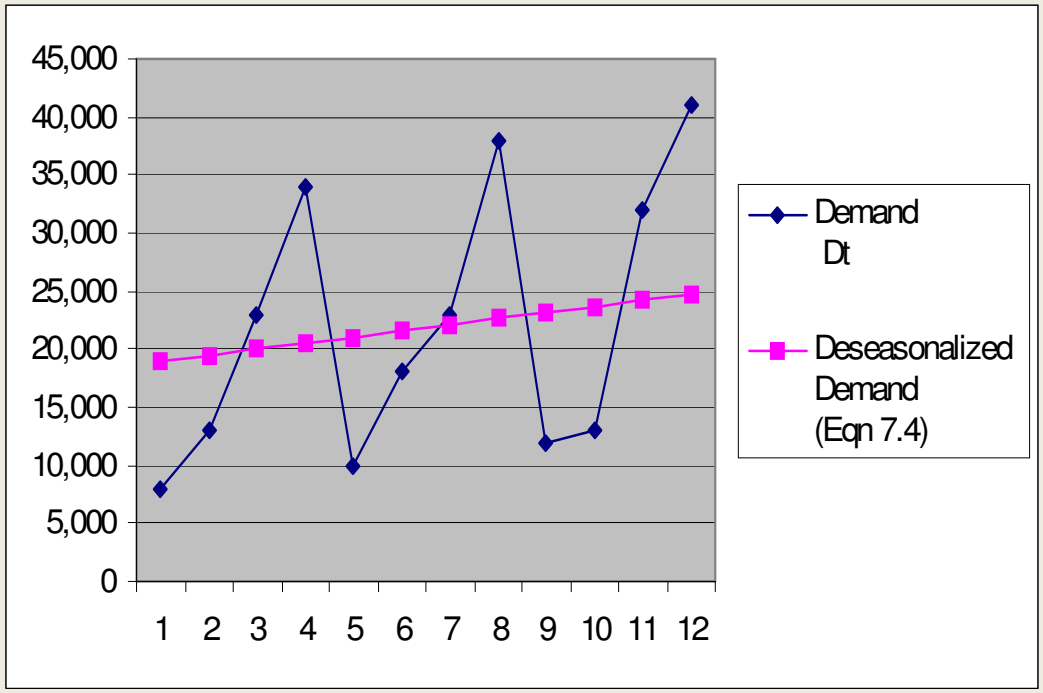
1	8,000	18,963
2	13,000	19,487
3	23,000	20,011
4	34,000	20,535
5	10,000	21,059
6	18,000	21,583
7	23,000	22,107
8	38,000	22,631
9	12,000	23,155
10	13,000	23,679
11	32,000	24,203
12	41,000	24,727

Period t

D_t

$$\overline{D_t} = 18439 + 524t$$

(Eqn. 7.4)



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Estimating Seasonal Factors

- Seasonal factor <u>for given period *t*</u> = ratio of actual demand to estimated deseasonalized demand :

$$\overline{S_t} = \frac{D_t}{\overline{\overline{D_t}}}$$

- Seasonal factor <u>for given season *i*</u>: average seasonal factors over periods that belong to that season

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Estimating seasonal factors

- Estimating seasonal factors for Tahoe Salt:

Period t	***Demand*** D_t	$\overline{D_t}$ ***Deseasonalized Demand (Eqn 7.4)***	$\overline{S_t}$ ***Seasonal Factor (Eqn 7.5)***	S_i ***Estimate (Eqn 7.6)***
1	8,000	18,963	0.42	0.47
2	13,000	19,487	0.67	0.68
3	23,000	20,011	1.15	1.17
4	34,000	20,535	1.66	1.67
5	10,000	21,059	0.47	
6	18,000	21,583	0.83	
7	23,000	22,107	1.04	
8	38,000	22,631	1.68	
9	12,000	23,155	0.52	
10	13,000	23,679	0.55	
11	32,000	24,203	1.32	
12	41,000	24,727	1.66	

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Determine forecast for future periods

- Forecast for period *k*:

$$F_k = (L + kT)S_i$$

i = season to which period *k* belongs

Forecasted Data		
Year, Qtr	***Period k***	***Forecasted Demand*** F_k
03,2	13	11,868
03,3	14	17,527
03,4	15	30,770
04,1	16	44,794
. . .	. . .	. . . .

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Adaptive forecasting

Adaptive Forecasting

→The estimates of level, trend, and seasonality are adjusted after each demand observation!!

- Moving average
- Simple exponential smoothing
- Trend-corrected exponential smoothing (Holt's model)
- Trend- and seasonality-corrected exponential smoothing (Winter's model)

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Moving Average

- Used when demand has no observable trend or seasonality
  - E.g.: TV sales
- Systematic component of demand = **level**
  - The estimate of the level at the end of period *t* is the average demand over the last N periods (the N-period moving average)

$$L_t = (D_t + D_{t-1} + \ldots + D_{t-N+1}) / N$$

  - Current forecast for all future periods is the same and is equal to the current estimate of the level

$$F_{t+1} = L_t \text{ and } F_{t+n} = L_t$$

  - After observing the demand for period t+1, revise the estimates as follows:

$$L_{t+1} = (D_{t+1} + D_t + \ldots + D_{t-N+2}) / N$$
$$F_{t+2} = L_{t+1} \text{ and } F_{t+n} = L_{t+1}$$

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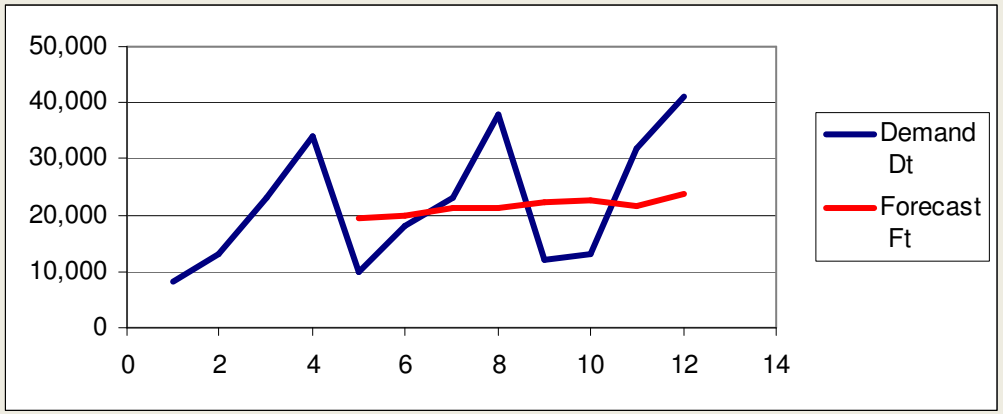
Moving Average for Tahoe Salt example (N=4)

Period t	Demand D_t	Level L_t	Forecast F_t	Error E_t
1	8,000			
2	13,000			
3	23,000			
4	34,000	19,500		
5	10,000	20,000	19,500	9,500
6	18,000	21,250	20,000	2,000
7	23,000	21,250	21,250	-1,750
8	38,000	22,250	21,250	-16,750
9	12,000	22,750	22,250	10,250
10	13,000	21,500	22,750	9,750
11	32,000	23,750	21,500	-10,500
12	41,000	24,500	23,750	-17,250
13			24,500	
14			24,500	
15			24,500	
16			24,500	



Forecast error: $E_t = F_t - D_t$

- How to choose N?
  - Different N: different error
- How to judge the quality of the method?

Estimate of level is revised every period

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Simple Exponential Smoothing

- Used when demand has no observable trend or seasonality

- Systematic component of demand = **level**
  - Initial estimate of level, L_0, assumed to be the average of all historical data

$$L_0 = \frac{\sum_{i=1}^{n} D_i}{n}$$

  - At the end of period *t*: current forecast for all future periods is the same and is based on the current estimate of the level

$$F_{t+1} = L_t \text{ and } F_{t+n} = L_t$$

  - After observing the demand for period t+1, revise the estimate as follows:

$$F_{t+2} = L_{t+1} = \alpha D_{t+1} + (1-\alpha)F_{t+1}$$

$0 \leq \alpha \leq 1$
smoothing constant

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Simple Exponential Smoothing

- Note:

$$F_{t+2} = L_{t+1} = \alpha D_{t+1} + (1-\alpha)F_{t+1}$$
$$= F_{t+1} + \alpha(D_{t+1} - F_{t+1})$$
$$= F_{t+1} - \alpha(F_{t+1} - D_{t+1})$$
$$= L_t - \alpha(E_{t+1})$$

E_{t+1} = Error in period t+1 = $F_{t+1} - D_{t+1}$

- Current estimate of level = Previous estimate of level + fraction of error that you made in the most recent period
- $\alpha = 0$: no correction for error
- as $\alpha \uparrow$: you make model more sensitive to error in the previous period

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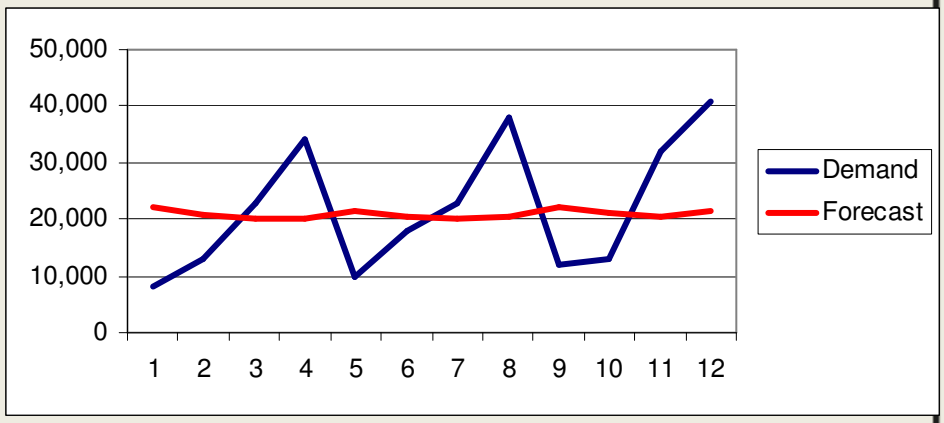
Simple Exponential Smoothing for Tahoe Salt example (α=0.1)

Period t	Demand D_t	Level L_t	Forecast F_t	Error E_t
0		22,083		
1	8,000	20,675	22,083	14,083
2	13,000	19,908	20,675	7,675
3	23,000	20,217	19,908	-3,093
4	34,000	21,595	20,217	-13,783
5	10,000	20,436	21,595	11,595
6	18,000	20,192	20,436	2,436
7	23,000	20,473	20,192	-2,808
8	38,000	22,226	20,473	-17,527
9	12,000	21,203	22,226	10,226
10	13,000	20,383	21,203	8,203
11	32,000	21,544	20,383	-11,617
12	41,000	23,490	21,544	-19,456
			23,490	
			23,490	
			23,490	
			23,490	



- How to choose α?
  - Different α : different error
- How to judge the quality of the method?

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Trend-Corrected Exponential Smoothing (Holt's Model)

Appropriate when the demand is assumed to have a level and trend in the systematic component, but no seasonality

- Obtain initial estimate of level and trend by running a linear regression in Excel

$$D_t = t.\ T_0 + L_0$$

- At the end of period t (t=1,…), the forecast for future periods is expressed as follows:

$$F_{t+1} = L_t + T_t$$
$$F_{t+n} = L_t + nT_t$$

- After observing demand for period t+1, revise the estimates for level and trend as follows:

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Trend-Corrected Exponential Smoothing (Holt's Model)

$$L_{t+1} = \alpha D_{t+1} + (1-\alpha)(L_t + T_t)$$
$$= \alpha D_{t+1} + (1-\alpha)F_{t+1}$$
$$= F_{t+1} - \alpha E_{t+1}$$

$$T_{t+1} = \beta(L_{t+1} - L_t) + (1-\beta)T_t$$

α = smoothing constant for level: $0 \leq \alpha \leq 1$
β = smoothing constant for trend: $0 \leq \beta \leq 1$

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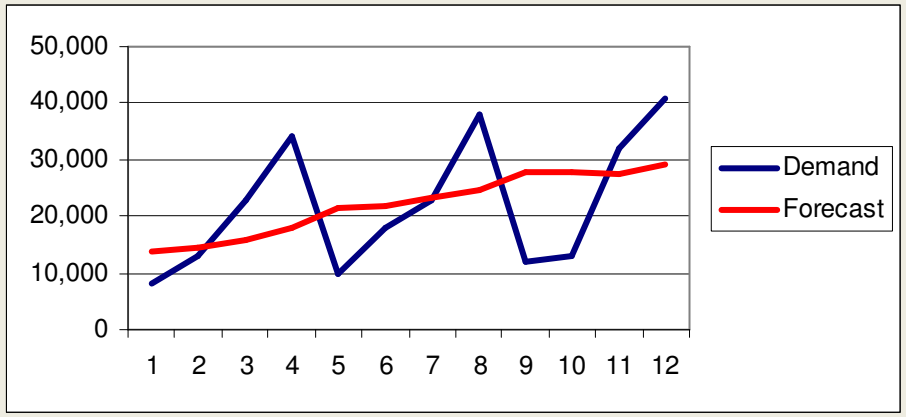
Holt's model for Tahoe Salt example (α=0.1 and β=0.2)

Period t	Demand D_t	Level L_t	Trend T_t	Forecast F_t	Error E_t
0		12,015	1,549		
1	8,000	13,008	1,438	13,564	5,564
2	13,000	14,301	1,409	14,445	1,445
3	23,000	16,439	1,555	15,710	-7,290
4	34,000	19,594	1,875	17,993	-16,007
5	10,000	20,322	1,645	21,469	11,469
6	18,000	21,570	1,566	21,967	3,967
7	23,000	23,123	1,563	23,137	137
8	38,000	26,018	1,830	24,686	-13,314
9	12,000	26,262	1,513	27,847	15,847
10	13,000	26,298	1,217	27,775	14,775
11	32,000	27,963	1,307	27,515	-4,485
12	41,000	30,443	1,541	29,270	-11,730
				31,985	
				33,526	
				35,067	
				36,609	



- How to choose α and β?
  - Different α, β: different error
- How to judge the quality of the method?

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Trend- and Seasonality-Corrected Exponential Smoothing (Winter's model)

Appropriate when the systematic component of demand is assumed to have a level, trend, and seasonal factor

- Systematic component = (level+trend)(seasonal factor)
- Assume periodicity p

- Obtain initial estimates of level (L_0), trend (T_0), seasonal factors ($S_1,\ldots,S_p$)
  - using procedure for static forecasting

- At the end of period t, the forecast for future periods is given by:

$$F_{t+1} = (L_t+T_t)(S_{t+1}) \text{ and } F_{t+n} = (L_t + nT_t)S_{t+n}$$

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Trend- and Seasonality-Corrected Exponential Smoothing (Winter's model)

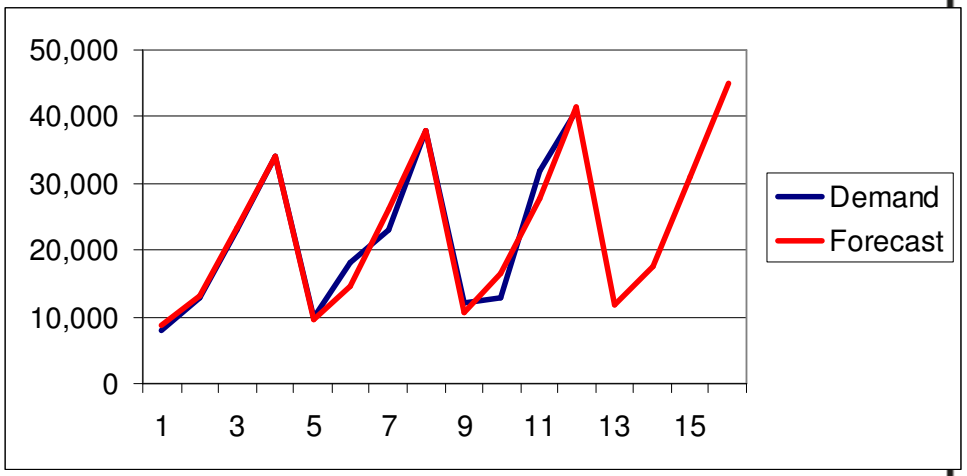
- After observing demand for period t+1, revise estimates for level, trend, and seasonal factors as follows:

$$L_{t+1} = \alpha(D_{t+1}/S_{t+1}) + (1- \alpha)(L_t+T_t)$$
$$T_{t+1} = \beta(L_{t+1} - L_t) + (1- \beta)T_t$$
$$S_{t+p+1} = \gamma(D_{t+1}/L_{t+1}) + (1- \gamma)S_{t+1}$$

α = smoothing constant for level: $0 \leq \alpha \leq 1$
β = smoothing constant for trend: $0 \leq \beta \leq 1$
γ = smoothing constant for seasonal factor: $0 \leq \gamma \leq 1$

Winter's model for Tahoe Salt example (α=0.05, β=0.1, γ=0.1)

Period t	Demand D_t	Level L_t	Trend T_t	Seasonal Factor S_t	Forecast F_t
		18,439	524		
1	8,000	18,866	514	0.47	8,913
2	13,000	19,367	513	0.68	13,179
3	23,000	19,869	512	1.17	23,260
4	34,000	20,380	512	1.67	34,036
5	10,000	20,921	515	0.47	9,723
6	18,000	21,689	540	0.68	14,558
7	23,000	22,102	527	1.17	25,981
8	38,000	22,636	528	1.67	37,787
9	12,000	23,291	541	0.47	10,810
10	13,000	23,577	515	0.69	16,544
11	32,000	24,271	533	1.16	27,849
12	41,000	24,791	532	1.67	41,442
13				0.47	11,940
14				0.68	17,579
15				1.17	30,930
16				1.67	44,928



- How to choose α, β and γ?
  - Different α, β, γ : different error
- How to judge the quality of the method?

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Measures of Forecast Error

- How can we find out which model is most appropriate?
  - How can we detect if we have trend? Seasonality?
  - How can we choose the "best values" for the parameters?

- Answer: measures of forecast error:
  - Forecast error should measure the RANDOM component of demand, if your method is appropriate
    - Any SYSTEMATIC error (consistent over- or underestimation, or pattern left) in the forecast indicates that you have an inappropriate forecasting method!!
    - Lower forecast error → higher accuracy → better model

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Measures of Forecast Error

- Forecast error for period *t*:

$$E_t = F_t - D_t$$

- Absolute deviation for period *t*:

$$A_t = \left| E_t \right|$$

- MSE (mean squared error):
  - very often used

$$MSE_n = \frac{1}{n} \sum_{t=1}^{n} E_t^2$$

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Measures of Forecast Error

- Mean absolute deviation:

$$MAD = \frac{1}{n}\sum_{t=1}^{n} A_t$$

- Mean absolute percentage error (MAPE):
  - Average absolute error as a percentage of demand

$$MAPE_n = \frac{\sum_{t=1}^{n}\left|\frac{E_t}{D_t}\right|100}{n}$$

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Measures of Forecast Error

- Bias:
  - To detect whether forecast consistently over- or underestimates true demand!!
  - Should fluctuate around zero and, ideally, be close to zero

$$bias_n = \sum_{t=1}^{n} E_t$$

- Tracking signal (TS):
  - Should be between -6 and +6, otherwise you are under- or overforecasting

$$TS_n = \frac{bias_n}{MAD_n}$$

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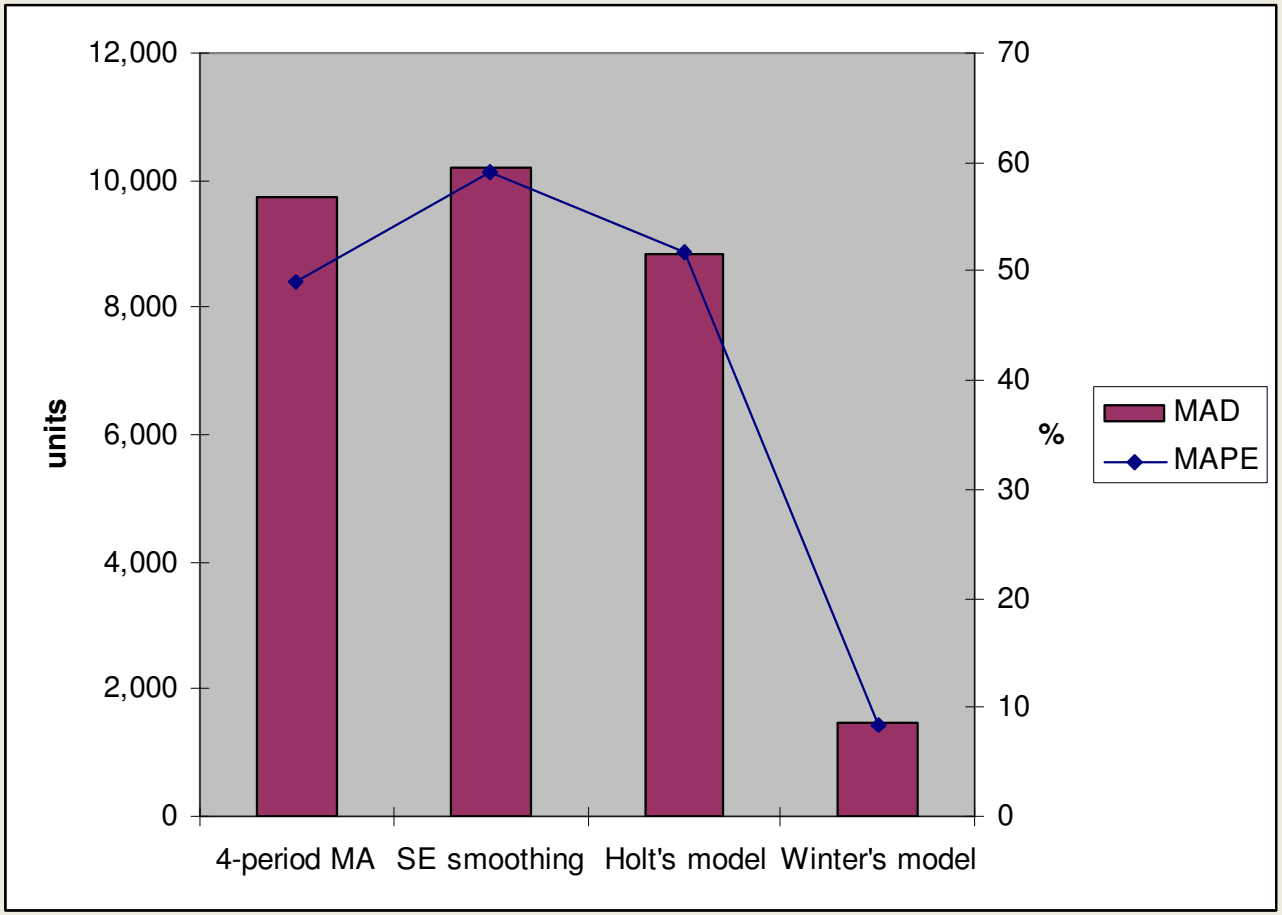
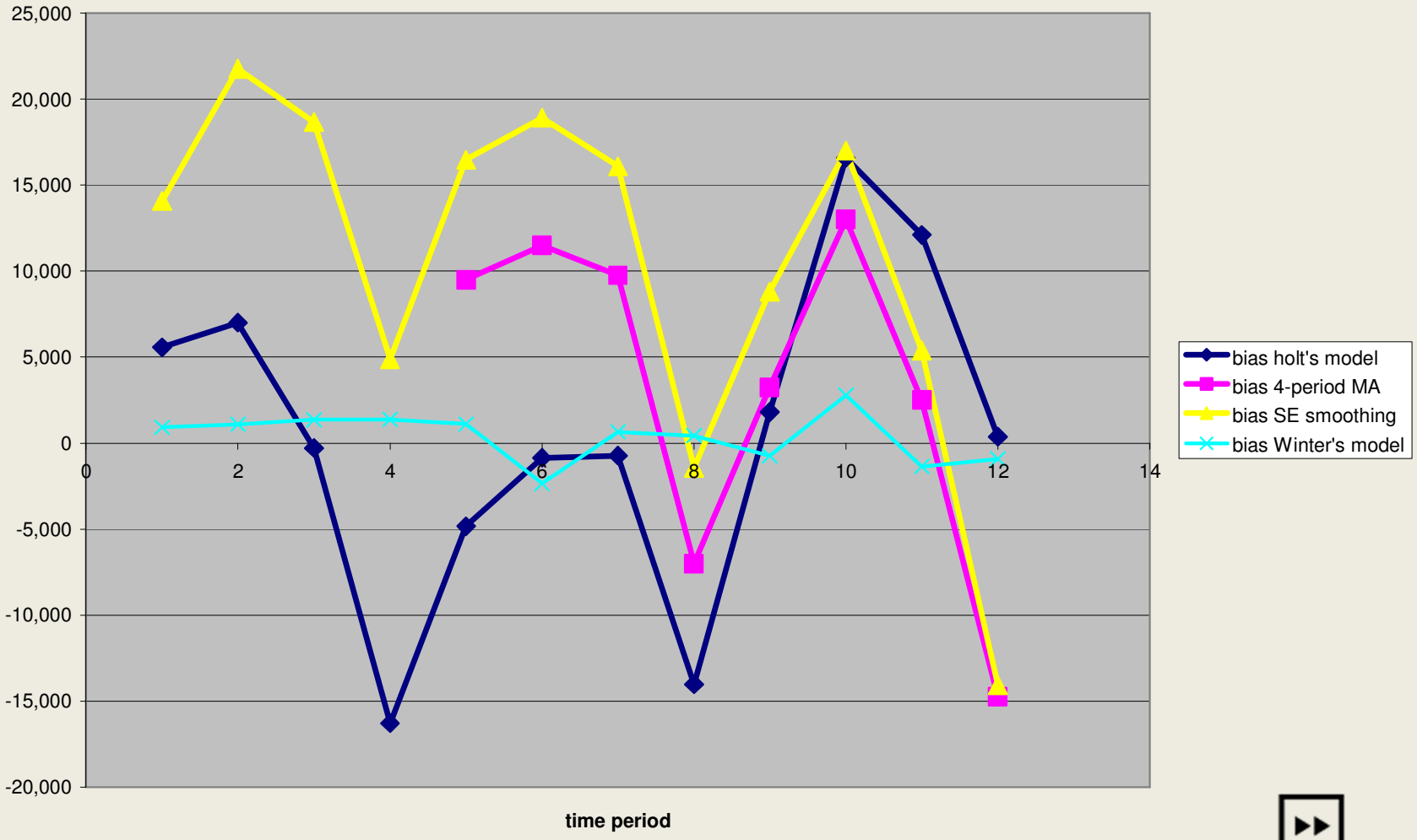
Measures of forecast error



Measures of Forecast Error



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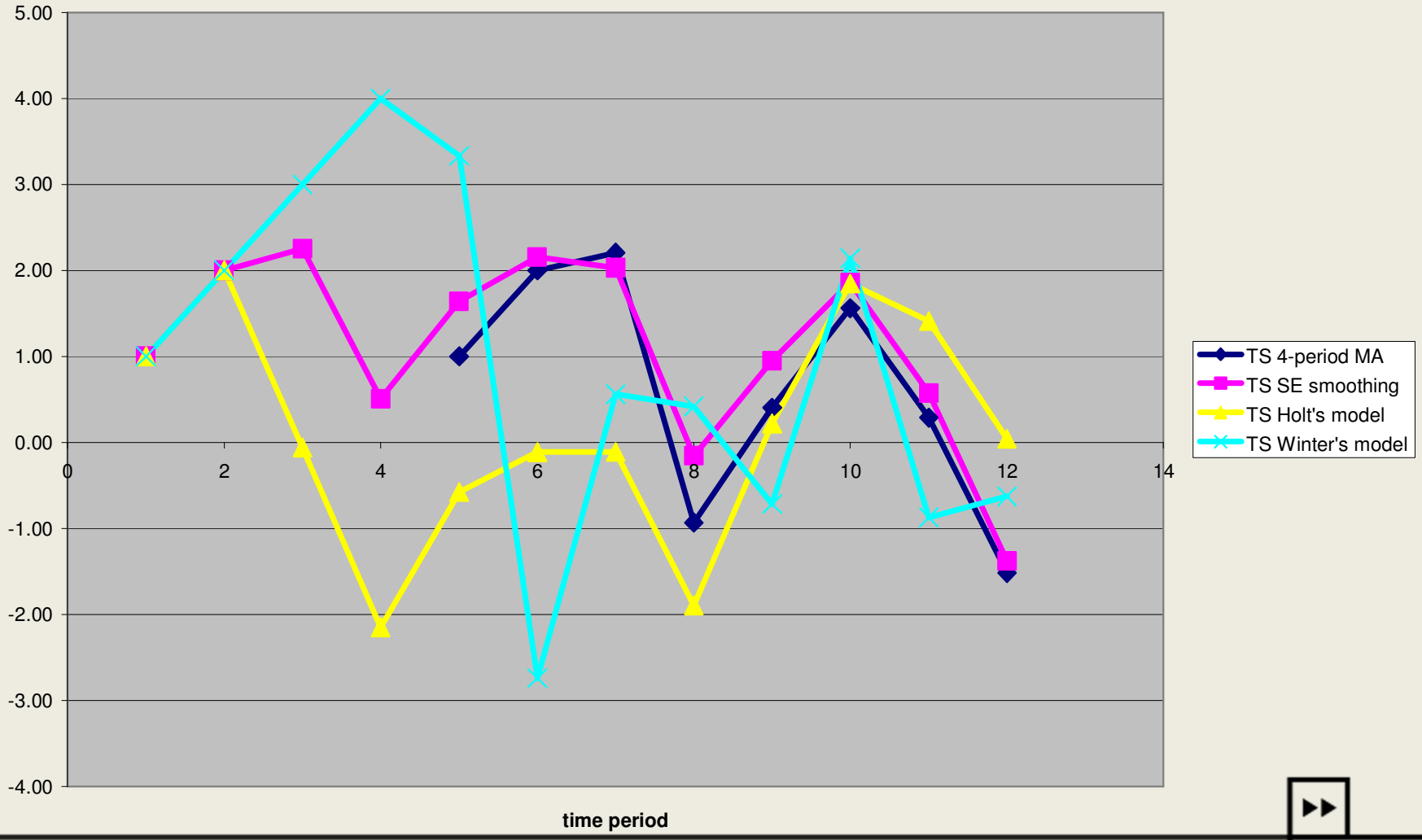
Measures of Forecast Error



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